ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PAPER 1 · PRACTICE D · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice d.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

## SECTION A

## QUESTION 1

**1.1** D ✓
**1.2** B ✓
**1.3** A ✓
**1.4** C ✓
**1.5** B ✓
**1.6** A ✓
**1.7** D ✓
**1.8** C ✓
**1.9** A ✓
**1.10** B ✓

(10)

**[10]**

## QUESTION 2

**2.1** True ✓
**2.2** False ✓
**2.3** True ✓
**2.4** True ✓
**2.5** False ✓
**2.6** True ✓
**2.7** True ✓
**2.8** False ✓
**2.9** True ✓
**2.10** False ✓

(10)

**[10]**

## QUESTION 3

**3.1** F ✓
**3.2** D ✓
**3.3** I ✓
**3.4** B ✓
**3.5** J ✓

**3.6** G ✓

**3.7** C ✓

**3.8** A ✓

**3.9** H ✓

**3.10** E ✓

(10)

**[10]**

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1** $4{,}5 \times 10^6$ ✓ (1)

**4.2** $0{,}0072$ ✓ (1)

**4.3** $a = (v - u) \div t$ ✓ $= (2 - 0) \div 8$ ✓✓ $= 0{,}25\ m/s^2$ ✓ (4)

**4.4** $F = m \times a = 5 \times 2$ ✓ $= 10\ N$ ✓ (2)

**4.5** $W = F \times s$ ✓ $= 50 \times 3$ ✓ $= 150\ J$ ✓ (3)

**4.6** $P = W \div t$ ✓ $= 150 \div 5$ ✓ $= 30\ W$ ✓ (3)

**4.7** Proton ✓ – positive ✓; neutron ✓ – no charge (neutral) ✓; electron ✓ – negative ✓ (6)

**[20]**

## QUESTION 5

**5.1.1** Polarised (electrolytic) capacitor ✓

**5.1.2** Potentiometer / variable resistor ✓

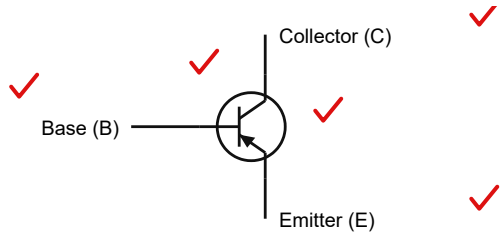
**5.1.3** Battery (of cells) ✓

**5.1.4** Ammeter ✓

**5.1.5** Earth (ground) ✓

(5)

**5.2**




(Base line 1, collector 1 and emitter 1 lines; arrow on the emitter pointing INWARDS (towards the base) 1; terminals labelled B, C, E 1)

(5)

**[10]**

## QUESTION 6

**6.1** A conductor has many free electrons and lets current flow easily ✓ – e.g. copper, aluminium, silver ✓. An insulator has very few free electrons and does not let current flow ✓ – e.g. rubber, PVC, glass, wood ✓. (4)

**6.2** $22 \times 1\,000 = 22\ \text{k}\Omega\ (22\,000\ \Omega)$ ✓✓; tolerance $\pm$ 10% ✓ (3)

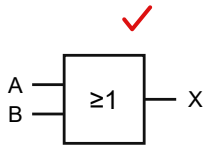
**6.3** $R_T = (R_1 \times R_2) \div (R_1 + R_2)$ ✓ $= (100 \times 300) \div 400$ ✓ $= 75\ \Omega$ ✓ (3)

**[10]**

## QUESTION 7

**7.1** $16 + 0 + 4 + 2 + 0$ ✓ $= 22$ ✓ (2)

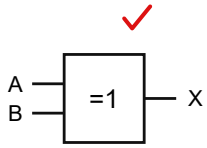
**7.2.1**




| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

(Correct IEC OR symbol: rectangle marked ≥1 1; truth table 4) (5)

**7.2.2**




| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(Correct IEC EX-OR symbol: rectangle marked =1 1; truth table 4) (5)

**7.3** An analogue signal changes smoothly and can have any value in a range ✓ (e.g. a sine wave from a microphone) ✓; a digital signal has only two levels, 0 and 1 (e.g. a square wave) ✓. (3)

**[15]**

## QUESTION 8

**8.1** Any THREE types ✓✓✓: water; foam; carbon dioxide ($CO_2$); dry chemical powder (DCP). Electrical fires: $CO_2$ or dry powder ✓ (4)

**8.2** Any FIVE in a sensible order ✓✓✓✓✓: do not touch the person while they are still in contact with the supply; switch off the power at the switch / breaker; if that is not possible, move the person away with a dry non-conducting object (wooden broom); call for help / emergency services; check breathing and pulse and start CPR if needed; treat burns with cool water and cover them; keep the person warm and still until help arrives. (5)

**8.3** Any THREE tools ✓✓✓ with their uses ✓✓✓, e.g. soldering iron – joins components with solder; side cutters – cut wires and component leads; wire strippers – remove insulation; long-nose pliers – hold and bend small parts; multimeter – measures voltage, current and resistance; desoldering pump – removes solder. (6)

**[15]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**